Topological Optimization Planar Elastic Energy-Storing Structures in a Fixed Area

Proposal

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Background & Motivation

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- TODO: Why is it significant? (fundamental interest / practical application)
- TODO: What does existing literature say, and what gap remains? [1]

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Project Goals

- **Goal 1:** Define a physically meaningful objective function for energy storage and output performance.
- **Goal 2:** Measure force displacement behavior to evaluate energy input, output, and efficiency.
- **Goal 3:** Analyze how topology affects load paths, deformation modes, and stiffness.

Hypotheses

- **H1:** Topologies with distributed deformation (e.g. serpentine or curved structures) will achieve higher energy storage efficiency than straight beam structures.
  - *Rationale:* More uniform deformation reduces stress concentration and energy loss.
- **H2:** Topology-optimized structures will outperform manually designed baselines in terms of energy efficiency and performance stability.
  - *Rationale:* Optimization explores non-intuitive designs that better balance stiffness and compliance.

Experimental Setup

Apparatus 3D printer, ,

Materials TODO: samples, reagents, consumables.

Instruments TODO: sensors, detectors, readout hardware.

Software Octave,

Environment TODO: temperature, pressure, shielding, etc.

Theoretical Basis

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- TODO: Key relation or equation, e.g.:

$$\text{TODO:} \quad f(x) = \dots$$

- TODO: Assumptions and their justification.

Analysis Plan

1. **Data acquisition:** TODO (what is recorded, at what rate/resolution)
2. **Preprocessing:** TODO (cleaning, calibration, normalisation)
3. **Feature extraction:** TODO (quantities to be computed)
4. **Statistical treatment:** TODO (error analysis, hypothesis tests)
5. **Validation:** TODO (controls, cross-checks, known benchmarks)

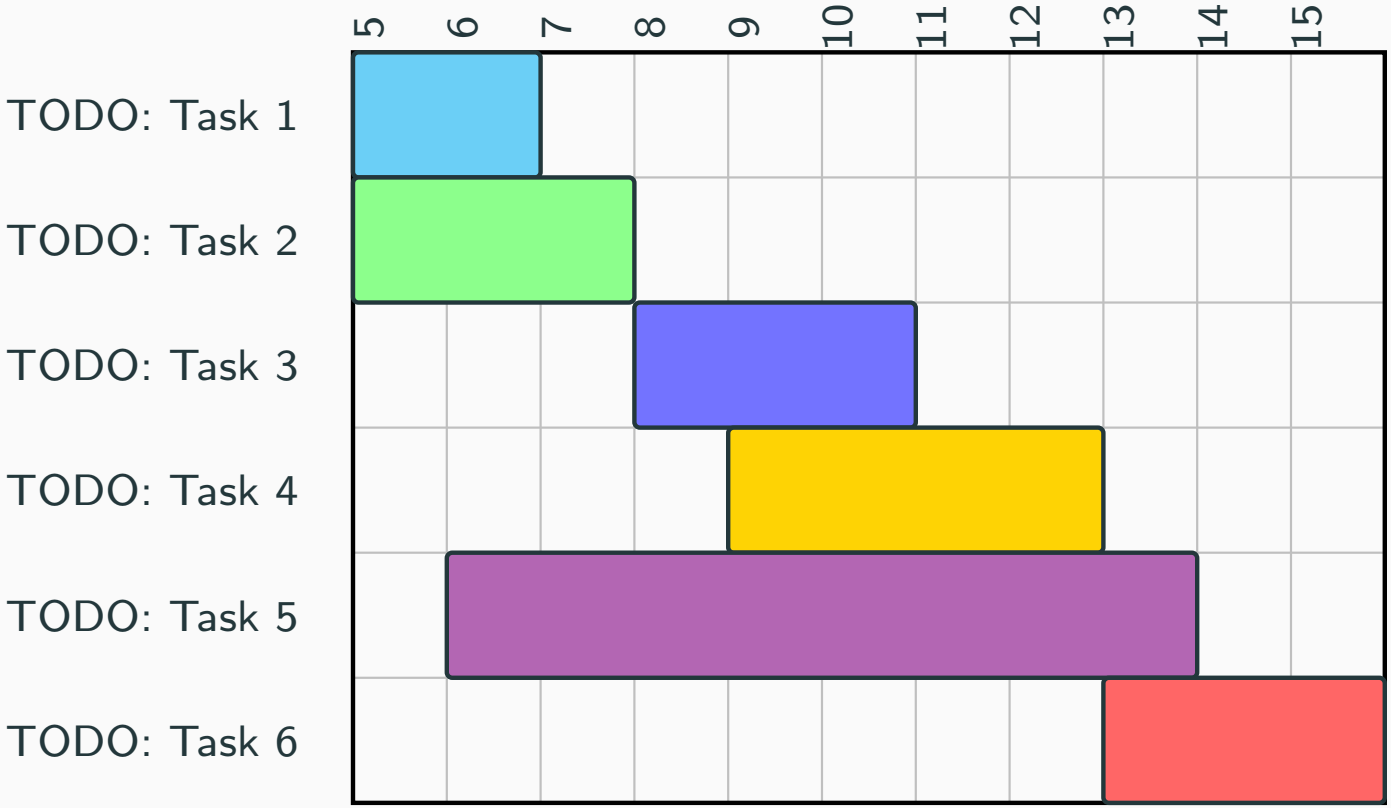
Expected Deliverables

- TODO: e.g. a characterised dataset / calibrated apparatus.
- TODO: e.g. a fitted model or extracted physical parameter.
- TODO: e.g. a comparison against theory or published values.

Safety & Compliance

- **Hazards identified:** TODO (chemical, electrical, radiological, ...)
- **Mitigation measures:** TODO (PPE, ventilation, interlocks, ...)
- **Regulatory status:** TODO
  - e.g. Not regulated under Act 304 / AELB because ...
- **Waste disposal:** TODO
- **Emergency contacts:** TODO (lab supervisor, safety officer)

Project Timeline



Budget

Item	Qty	Unit (MYR)	Total (MYR)
Equipment			
TODO: Item name	TODO	TODO	TODO
TODO: Item name	TODO	TODO	TODO
Materials & Consumables			
TODO: Item name	TODO	TODO	TODO
TODO: Item name	TODO	TODO	TODO
Software & Licences			
Open-source only	–	–	0
Miscellaneous			
TODO: Item name	TODO	TODO	TODO
Total			**TODO**

TODO: e.g. Requested from Physics Dept. student project fund (cap: MYR 500).

References i

[1] F. Last, "Paper title," *Journal Name*, vol. 1, pp. 1–10, 2024. DOI: 10.xxxx/xxxxx